higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 2

(6050372)

1 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Paper can be used five times before fibre becomes too short to form paper.

1.2 A fluidised bed reactor does not emit smelly gas.

1.3 Calcium bisulphite is used as cooking liquor in the NSSC process.

1.4 The primary function of the drying section is to consolidate the sheet of paper.

1.5 Embossed papers are used for any paper product where a decorative effect is required e.g. greeting cards.

(5 × 1) **[5]**

QUESTION 2

Give ONE word/term for each of the following descriptions. Write only the word/term next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 The removal of unsuitable chips and contraries from the product.

2.2 Used cooking liquor from kraft cook.

2.3 A viscous yellow-black odorous liquid obtained as a co-product of the kraft process.

2.4 The operation which reduces the continuous web or webs of paper coming from the rolls to specified sizes of separate sheet.

2.5 A place of intersection where one roll touches another.

(5 × 1) **[5]**

QUESTION 3

Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–H) next to the question number (3.1–3.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1	Wavy inside boxes	A	chemical recovery
3.2	The breaking down of fibrous materials	B	digestion
3.3	A process that converts the chemicals produced by the digestion process back into cooking liquor	C	fluting
		D	lithography
3.4	A process whereby rolls are prepared for shipment	E	bowls
		F	carton boards
3.5	Printing process reliant on the fact that oil and water do not mix	G	roll finishing
		H	pulping

(5 × 1) **[5]**

QUESTION 4

4.1 Paper has been used since 140 BC as a form of written communication.

Who is regarded as the inventor of paper? (1)

4.2 List THREE Sappi divisions. (3)

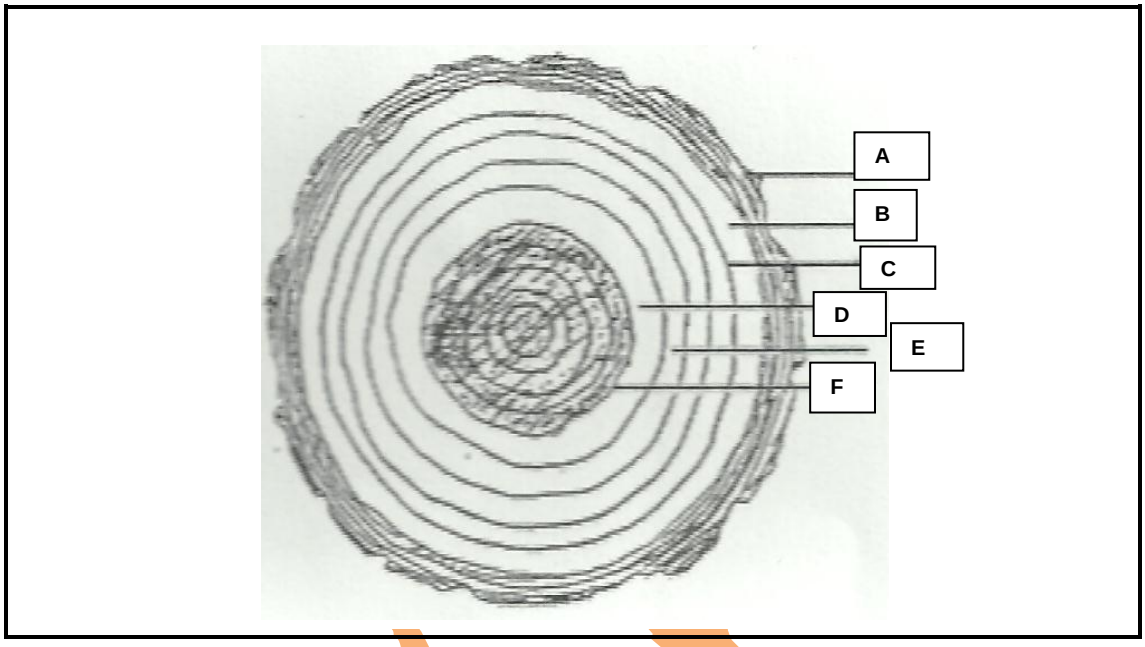
4.3 Pulp and paper mills generate a lot of solid waste in the form of sludge that may contain contaminants that can be dangerous to humans.

What is the main method of disposing solid waste products? (1)

4.4 Name THREE criteria that are used to determine the quality of waste paper. (3)

4.5 Define *contraries*. (2)

4.6 A diagram below represents a cross section of a stem of a tree trunk. Label the diagram and write only the answer next to the question number (A–F) in the ANSWER BOOK.



(6)

4.7 Name THREE factors that can affect the quality of bagasse. (3)
[19]

QUESTION 5

5.1 Describe how the moisture content of paper can be measured on an air dry basis. (4)

5.2 Define a *tissue paper*. (1)

5.3 Name FOUR uses of tissue paper. (4)

5.4 Bark contains particles that can affect the quality of pulp if not removed.

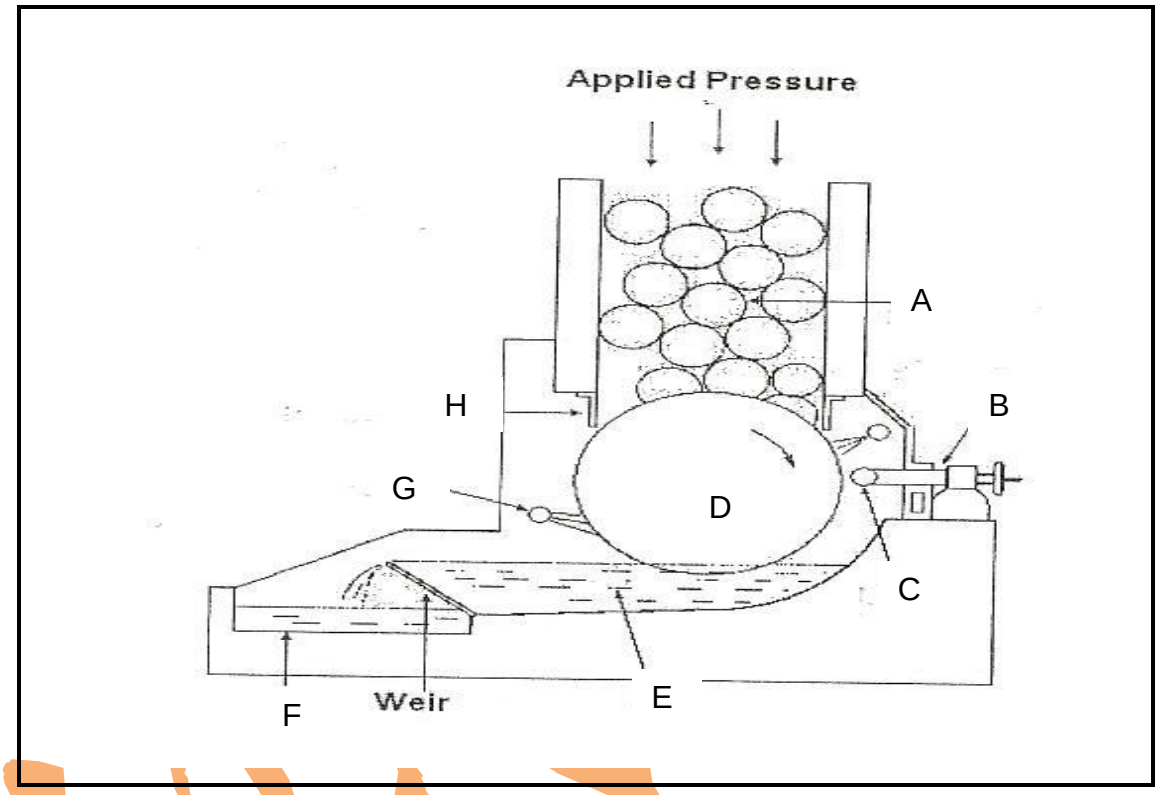
5.4.1 Give FOUR reasons why it is important to remove bark from logs. (4)

5.4.2 What is the most common equipment used to remove bark from logs? (1)

5.5 Name kinds of contaminants found in chips. (4)
[18]

QUESTION 6

6.1 Complete the diagram below of a grinder used for ground wood pulping by naming components (A–H) in the ANSWER BOOK.



(8)

6.2 Name TWO chemicals that can be used in the kraft process. (2)

6.3 Name TWO types of digesters used in the kraft process. (2)
[12]

QUESTION 7

7.1 Name FOUR preventative steps to be taken to minimise formation of sulphur trioxide (SO_3) during calcium bisulphite production. (4)

7.2 Sketch a labelled flow diagram of the lime burning process used in the production of calcium bisulphite. (8)
[12]

QUESTION 8

8.1 Name FIVE advantages of using hybrid formers. (5)

8.2 There are four main phases in the pressing process.

8.2.1 Briefly describe phase 1 of the pressing process. (6)

8.2.2 Define *pressing.* (2)

[13]

QUESTION 9

9.1 Roll finishing is the process in which rolls are prepared for shipment.

9.1.1 Name and describe THREE steps in roll finishing. (6)

9.1.2 Name TWO processes involved in finishing operations. (2)

9.2 Briefly explain how one can rewind a roll of paper to the customer's requirements if it cannot be done by the machine winder. (3)

[11]

TOTAL: 100

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